

ds 55 presentation

Team members:

Timeline / Content

1. Problem to Present – Objective - Maria
2. Models to Explore - Lydia
3. Performance Comparison by Model
 - Linear - Jorge
 - Random Forest / Decision Tree - Maverick
 - Light GBM - Luoxu
 - XGBoost - Riva
4. **Performance by Price Band - Sammy**
5. **Stability over time - Luoxu**
6. **Iteration Timeline – Riva**
7. **What work - Kelvin**
8. **what didn't work - Kelvin**

Objective

Our objective was to develop a machine learning (ML) model to accurately predict the closing price of single-family residential properties using the California Regional Multiple Listing Service (CRMLS) data.

Input: Property and listing characteristics

Model: Various ML regression and tree models

Output: Predicted Close Price

We wanted to answer the question of how accurately we can predict final sale price with the information we have?

Dataset

Dataset Source: California Regional Multiple Listing Services (CRMLS)

Original time period: May 2025 – May 2026

Final time period: June 2025 – June 2026

Scope: Residential Properties, Single Family residences, California

Target Variable: ClosePrice

Initial Data Exploration

- We conducted EDA Analysis to measure how we could go about using the data. Some of our key findings included:
 - Strongly right-skewed ClosePrice
 - LotSize & LivingArea were also strongly right-skewed
 - Some features had extreme outliers
 - Missing variables throughout the data

Dataset Challenges

- **Missing Data:** Incomplete property characteristics across various listings. Each listing had at least one missing value.
- **Extreme Values:** Usually high or low property measurements
- **Skewed Distributions:** Such as ClosePrice, LivingArea, LotSize, and other numerical values.
- **High Volume Data:** Large number of both numerical and categorical variables

Preprocessing Pipeline

- Idea for slide: we can try and make it like a diagram type of style slide and for the actual presentation we can go more into details.
- We can talk about how we filter and clean data, and how we handles things like outliers and also used log transformation for our target and some other variables. And so on

Feature Engineering

Models Used:

Model Comparison

Best Model

- List each member's best-performing model

Performance By Price Band

- [Can plot a distribution of the group's results]
- Riva's XGBoost model

Evaluation

- [Can plot a distribution of the group's results]

Conclusions